

JEE MAIN REVISION MASTER: S-BLOCK ELEMENTS OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	S Block Elements - Core Application	3 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 S Block Elements - Core Application (Moderate Yield — 3 PYQs)

Core Principle: Key problem pattern in S Block Elements.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2023 Evening Shift] Q#1 (Numerical): "The number of alkali metal(s), from Li,K,Cs , Rb having ionization enthalpy greater than 400kJ mol^{-1} and forming stable super oxide..." [Ans: 2]
- [2023 Evening Shift] Q#2 (Numerical): "On heating, LiNO_3 gives how many compounds among the following _____ Li_2O , N_2O , O_2 , LiNO_2 , NO_2 ..." [Ans: 3]
- [2022 Evening Shift] Q#3 (Numerical): " BeO reacts with HF in presence of ammonia to give [A] which on thermal decomposition produces [B] and ammonium fluoride. Oxidation state of ..." [Ans: 2]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 31st January Evening Shift (Q#1)

The number of alkali metal(s), from Li,K,Cs , Rb having ionization enthalpy greater than 400kJ mol^{-1} and forming stable super oxide is _____.

Correct Answer: 2

Solution Analysis: Li does not form superoxide. K, Rb and Cs form stable super oxides but Cs has ionisation enthalpy less than 400 kJ . While K and Rb have ionisation enthalpy greater than 400 kJ mol^{-1} .